Exercise 19 – Spring Data JPA Quick Example

Aim

The aim of this exercise is to create a simple Spring Boot application using Spring Data JPA and Hibernate to access data from a MySQL database.

Description

In this exercise, a Spring Boot project named **orm-learn** was created using Maven. Spring Data JPA, Spring Web, MySQL Driver, and DevTools dependencies were added to the project.

A **Country** entity class was created to represent the country table. A **CountryRepository** interface extending JpaRepository was used to perform database operations. A **CountryService** class was created to retrieve the list of countries, and the **OrmLearnApplication** class was used to display the data.

The database configuration was added in the application.properties file, and the SQL script was prepared to create the database, table, and sample records.

Files Created

- Country.java
- CountryRepository.java
- CountryService.java
- OrmLearnApplication.java
- application.properties
- country.sql

SQL Script

CREATE DATABASE ormlearn;

USE ormlearn;

CREATE TABLE country (

```sql
    co_code VARCHAR(2) PRIMARY KEY,
    co_name VARCHAR(50)
);

INSERT INTO country VALUES
('IN','India'),
('US','United States'),
('JP','Japan'),
('FR','France'),
('DE','Germany');
```

Expected Output

Countries:

Country{coCode='IN', coName='India'}

Country{coCode='US', coName='United States'}

Country{coCode='JP', coName='Japan'}

Country{coCode='FR', coName='France'}

Country{coCode='DE', coName='Germany'}

Result

The Spring Boot project was created successfully. Spring Data JPA and Hibernate were configured, and the application was able to demonstrate how country data can be accessed using the repository and service classes.